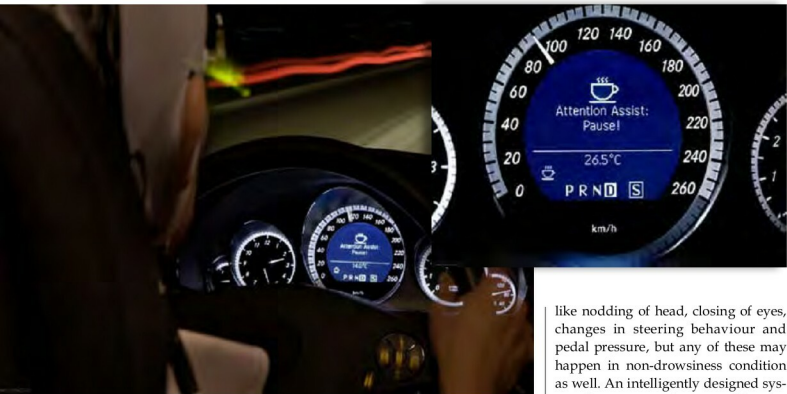


NEW CARS WON'T LET YOU SLEEP!

A study shows that a quarter of all serious road accidents are caused by drowsy drivers. The all new Mercedes-Benz E-Class car comes with an innovative and intelligent electronic system, Attention Assist, which detects drowsiness and alerts you to take a break when you feel sleepy while driving



Attention Assist technology in Mercedes-Benz E-Class

■ SHWETA DHADIWAL BAID

Today, automobile engineering is not just about mechanical sub-systems. About 40 per cent of a vehicle's cost involves integrated electronic systems. Power steering, power brakes and power windows, which were originally mechanical or based on hydraulic systems, have now given way to electronic control systems. Similarly, electronics plays a significant role in car safety systems. A variety of sensors and microcontroller units work in unison to give you the advantage of 'safe drive.'

Thanks to the development in sensor technology and information processing, the new Mercedes-Benz E-class features 'Attention Assist'—an intelligent electronic system that detects the driver's level of attention and warns him of drowsiness well in time. Drowsiness has been a primary safety concern for many automobile manufacturers. Many designers have come up with different techniques to detect drowsiness and fatigue and warn the driver through some audio/visual system.

In theory, drowsiness or fatigue can be detected by numerous parameters

like nodding of head, closing of eyes, changes in steering behaviour and pedal pressure, but any of these may happen in non-drowsiness condition as well. An intelligently designed system that may monitor a combination of parameters can really be helpful in ensuring safety of the car occupants.

The consequence of even one-second nap can be disastrous—at any speed and in any driving condition. "In order to detect drowsiness, Mercedes experimented with cameras in the beginning, yet discovered that the results were not according to the set standards," explains Dr Joerg Breuer, head of active safety, Daimler AG. "In extensive tests, the steering behaviour proved reliable and early indicator of the driver's level of alertness," he adds.

Steering wheel sensor

There are several methods to detect